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Overview

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Goals and Research Question

Base Agent

- combination of Behavioral Cloning (BC) and Proximal Policy Optimization (PPO)
- strong baseline, but weaknesses against several opponents

Goal

- improve the Base Agent with League Training (LT)
- training against a more diverse set of opponents
- focus on robustness and generalization

Research Question

- How much does LT with PPO improve robustness and generalization of the BC/PPO-pretrained agent?
- How strongly does this depend on the diversity of initial agents and BC experts?

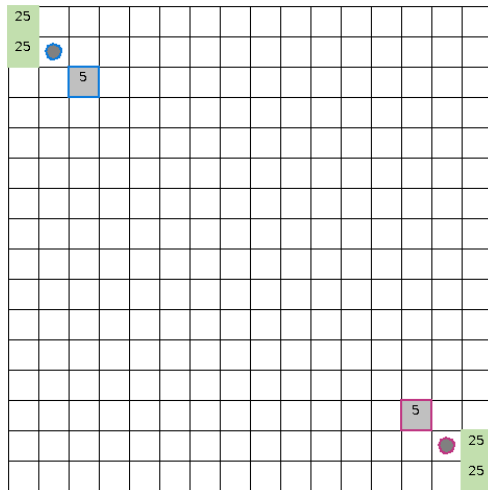
Setting

MicroRTS

- simplified RTS environment for RL research
- still contains resource management, unit production and combat
- supports controlled and repeatable experiments

Main RL challenges

- large combinatorial action space
- simultaneous decisions
- delayed and sparse rewards
- robustness against diverse opponents



Click the play area in Foxit Reader to start the embedded MicroRTS example video.

BC $\rightarrow$ BC-FT $\rightarrow$ PPO $\rightarrow$ LT $\rightarrow$ PPO fine-tuning

- **BC / BC-FT:** imitate strong scripted opponents to obtain a competent initial policy
- **PPO:** refine the policy through exploration
- **LT:** train against a population of evolving opponents instead of only bots or pure self-play
- **Fine-tuning:** fix remaining weaknesses after LT, especially against WorkerRushAI

From SP to LT

- SP** train against the current version of the agent; simple, but prone to overfitting or cyclical strategies
- FSP** train against a set of past policies; reduces forgetting, but opponent choice is still crude
- PFSP** sample opponents based on current win rates; focuses on informative and non-trivial matchups
- LT** add role-based agents: Main Agent, Historical Agents, Main Exploiters and League Exploiters

Why LT?

LT does not only preserve diversity through historical snapshots, it also creates diversity through exploiters that actively search for weaknesses of the current policy.

League Training Design

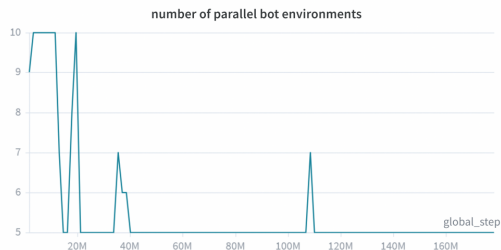
- **Main Agent:** primary policy
- **Historical Agents:** frozen snapshots for stability
- **Main Exploiters:** exploit weaknesses of historical Main Agents
- **League Exploiters:** search for broader league weaknesses

Key idea

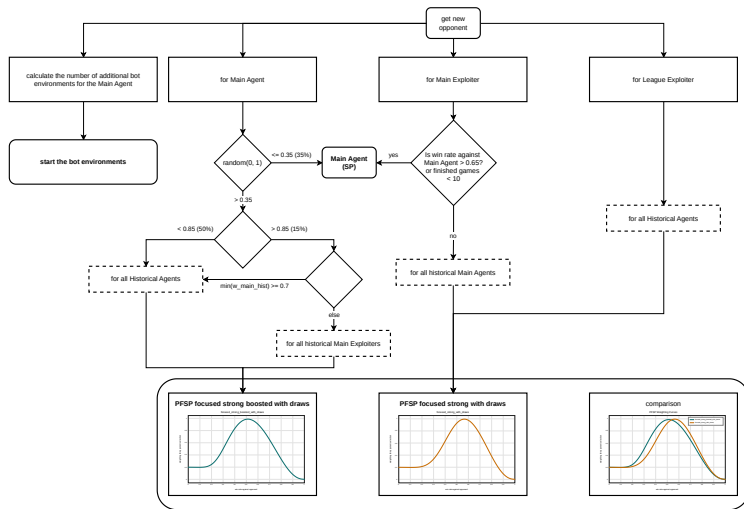
LT turns robustness against diverse opponents into part of the optimization target.

Bot Environments

- additional dynamic bot environments (CoacAI and Mayari)
- scripted bots: CoacAI, Mayari, WorkerRushAI, PassiveAI, LightRushAI, RandomAI and RandomBiasedAI
- MCTS based bots: Rojo, MixedBot, Izanagi, Tiamat, Droplet, GuidedRojoA3N and NaiveMCTS



LT Matchmaking



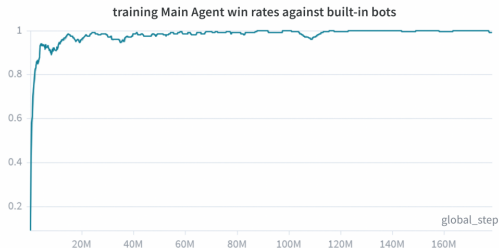
Matchmaking and opponent sampling in the LT setup.

Other Implementation Changes

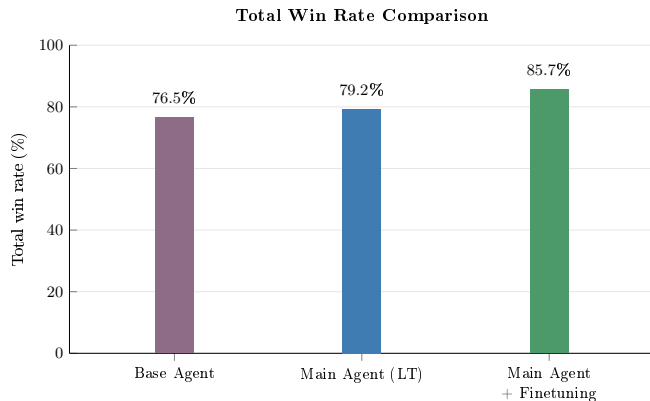
- delta score
- annealed entropy coefficient
- new initial agents

Experiment Setup

- map: basesWorkers16x16A
- LT setup: 1 Main Agent, 4 Main Exploiters, 1 League Exploiter
- 25 environments per learning agent
- 5 to 10 additional dynamic bot environments
- approximately 130 million LT steps
- training time: about 100 hours
- maximum setup size: 160 parallel environments
- evaluations over at least 100 games against 12 opponents after LT and after fine-tuning



Main Quantitative Result



- overall loss rate drops from **7.7%** to **2.2%**

Evaluation Results

Opponent	Main Agent without new initial agents	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
<i>bots used in training</i>				
CoacAI	100.0	96.0	100.0	100.0
Mayari	100.0	67.3	100.0	100.0
<i>Built-in bots</i>				
WorkerRushAI	31.0	100.0	35.0	99.5
PassiveAI	100.0	99.7	100.0	100.0
LightRushAI	100.0	98.0	100.0	100.0
RandomAI	100.0	99.0	100.0	100.0
RandomBiasedAI	91.0	79.0	97.0	97.5
<i>Other bots</i>				
Rojo	100.0	81.0	100.0	100.0
MixedBot	45.0	43.3	57.0	60.5
Izanagi	68.0	71.7	70.0	72.5
Tiamat	94.0	93.0	99.0	99.0
Droplet	45.0	67.7	62.0	75.5
GuidedRojoA3N	64.0	74.7	87.0	88.5
NaiveMCTS	0.0	0.7	2.0	6.5
overall	74.1	76.5	79.2	85.7

Evaluation Results

Opponent	Main Agent without new initial agents	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
<i>bots used in training</i>				
CoacAI	0.0	2.0	0.0	0.0
Mayari	0.0	27.3	0.0	0.0
<i>Built-in bots</i>				
WorkerRushAI	63.0	0.0	55.0	0.5
PassiveAI	0.0	0.0	0.0	0.0
LightRushAI	0.0	2.0	0.0	0.0
RandomAI	0.0	0.0	0.0	0.0
RandomBiasedAI	0.0	0.0	0.0	0.0
<i>Other bots</i>				
Rojo	0.0	5.3	0.0	0.0
MixedBot	33.0	34.3	20.0	13.5
Izanagi	15.0	16.3	7.0	6.0
Tiamat	6.0	7.0	1.0	0.5
Droplet	22.0	9.0	22.0	10.5
GuidedRojoA3N	0.0	1.7	0.0	0.0
NaiveMCTS	0.0	2.7	0.0	0.0
overall	9.9	7.7	7.5	2.2

Interpretation of the Results

hypothesis:

- LT turns robustness against diverse opponents into part of the optimization target
- LT helps to find a better local optimum that is then further improved by fine-tuning, while pure PPO often gets stuck in not as generalized local optima.

Player 1 Priority

Observation

MicroRTS gives player 1 action priority, so a potential concern is that self-play style training could accidentally favor player-1-specific behavior.

Result from the thesis

In one self-match evaluation of the same agent over 200 games, player 1 achieved 38% wins, 28% draws and player 2 achieved 34% wins.

- minimal effect on evaluations
- alternating the learning side to prevent the agent from exploiting player-1-specific priority

[▶ Open Player 1 Priority Example Video](#)

Base Agent vs Base Agent

Video example for the Base Agent's preferred strategy

▶ Open Video

Main Limitation: Diversity

- Exploiters initialized from similar policies tend to discover similar weaknesses.
- Some strategically different exploits are hard to reach because they require first unlearning the Base Agent's preferred behavior.
- New initial agents with focuses on different unit types to improve exploiter diversity and Main Agent generalization.
- Hypothesis: LT therefore depends strongly on the diversity already present at initialization

Scope and Remaining Limitations

- evaluation is limited to one map: basesWorkers16x16A
- the setting uses full observability and no fog of war
- high computational cost
- exploiter diversity

Takeaway

The method works, but the strength of the final result depends strongly on the diversity and cost of the training setup.

Future Work

- train on multiple maps and with fog of war
- improve the diversity and strength of initial agents
- reduce computational cost of MicroRTS training
- investigate stronger diversity mechanisms for exploiters

Conclusion

Conclusion

LT with PPO improves the robustness and generalization of the MicroRTS Base Agent. Additional PPO fine-tuning improves the final agent even further.

Key takeaway

The effectiveness of LT depends strongly on exploiter diversity, which in turn depends on the diversity of the initial agents and BC experts.

- LT improves most matchups and the overall performance
- fine-tuning further improves the final agent
- diversity of initial agents is the critical practical factor

Thank you

Questions?